

Code-Layout, Readability and Reusability

Date	15 March 2026
Team ID	SWTID-2026-5650
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	PEP 8 formatting enforced across python codebase.
2	Proper File Structure	Files and folders are logically organized	Yes	Structured into app/, components/, utils/, and prompts/.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive names (e.g. comic_script, panel_prompt).
4	Function / Method Names	Functions are descriptively named	Yes	Clear verb-noun naming (e.g. generate_comic_panel()).
5	Code Comments	Inline and block comments explain logic	Yes	Comprehensive docstrings and inline prompt notes.
6	Modular Design	Code is split into reusable functions/modules	Yes	Decoupled Gemini story engine, art, and layout renderer.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Shared utility functions for API buffers and PDF creation.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Try-except blocks covering API rate limits and render errors.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	gemini_story_engine.py	Python / Gemini 1.5 Pro	Multi-panel script generation and dialogue drafting	High

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
2	panel_image_generator.py	Python / Imagen 3 / Vision	Panel illustration pipeline and style consistency	High
3	comic_layout_renderer.py	Python / PIL / ReportLab	Rendering panel grids, speech bubbles, and PDF export	High
4	prompt_template_builder.py	Python / Jinja2 / LangChain	Formatting character sheets and panel style prompts	High
5	session_state_manager.py	Python / Streamlit State	Preserving user story state and panel edit history	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Clean directory hierarchy with complete separation of concerns.
Readability	5	Self-documenting code complying with PEP 8 standards and inline notes.
Reusability	5	Modular architecture with decoupled AI story and image creation modules.
Documentation / Comments	5	Full function docstrings, setup README, and code annotations.
Overall Score	5	Exceptional code quality meeting production-grade standards.